

Advanced Functions & Equations

ANSWER KEY



Function composition

- 1 Let $f(x) = 3x - 2$ and $g(x) = x^2 + 1$. Find $f(g(2))$ and $g(f(2))$.
- $g(2) = 5$, so $f(g(2)) = 3(5) - 2 = 13$. $f(2) = 4$, so $g(f(2)) = 4^2 + 1 = 17$.
- 2 Using the same f and g , find the formula for $(f \circ g)(x)$.
- $(f \circ g)(x) = f(g(x)) = 3(x^2 + 1) - 2 = 3x^2 + 1$.
-

Inverse functions

- 3 Find $f^{-1}(x)$ for $f(x) = 3x - 2$.
- Let $y = 3x - 2$. Solve for x : $x = \frac{y+2}{3}$. So $f^{-1}(x) = \frac{x+2}{3}$.
- 4 Verify your answer above by computing $f(f^{-1}(5))$.
- $f^{-1}(5) = \frac{5+2}{3} = \frac{7}{3}$, and $f\left(\frac{7}{3}\right) = 3\left(\frac{7}{3}\right) - 2 = 7 - 2 = 5$. It matches, confirming the inverse.
-

Rational equations

- 5 Solve $\frac{2x-1}{x+3} = \frac{1}{2}$.
- Cross-multiply: $2(2x-1) = x+3 \Rightarrow 4x-2 = x+3 \Rightarrow 3x = 5 \Rightarrow x = \frac{5}{3}$ (and $x \neq -3$, so this is valid).
- 6 Solve $\frac{3}{x-1} + 2 = 5$.
- $\frac{3}{x-1} = 3 \Rightarrow x-1 = 1 \Rightarrow x = 2$ (and $x \neq 1$, so this is valid).
-

Factoring higher-degree polynomials

- 7 Given that $x = 1$ is a root of $x^3 - 2x^2 - 5x + 6 = 0$, use the Factor Theorem to factor completely and find all roots.
- By the Factor Theorem, $(x - 1)$ is a factor. Dividing gives
- $x^3 - 2x^2 - 5x + 6 = (x - 1)(x^2 - x - 6) = (x - 1)(x - 3)(x + 2)$. Roots: $x = 1, 3, -2$.
- 8 Verify that $x = -2$ is a root of $x^3 - 2x^2 - 5x + 6 = 0$ by direct substitution.
- $(-2)^3 - 2(-2)^2 - 5(-2) + 6 = -8 - 8 + 10 + 6 = 0$. Yes, $x = -2$ is a root.

Exponential and logarithmic equations

9 Solve $3^{x-1} = 81$.

$$81 = 3^4, \text{ so } x - 1 = 4 \Rightarrow x = 5.$$

10 Solve $\log_3(x+2) + \log_3(x-2) = 2$, checking for extraneous roots.

Combine: $\log_3[(x+2)(x-2)] = 2 \Rightarrow x^2 - 4 = 3^2 = 9 \Rightarrow x^2 = 13 \Rightarrow x = \pm\sqrt{13}$. The domain requires $x - 2 > 0$, i.e. $x > 2$; since $\sqrt{13} \approx 3.61 > 2$ but $-\sqrt{13} < 2$, only $x = \sqrt{13}$ is valid.